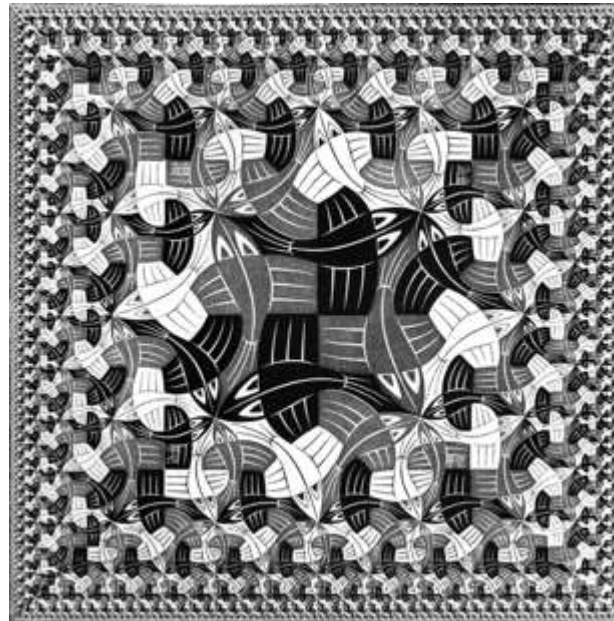


Functional geometry

Picture combinators & recursive fish

Einar W. Høst



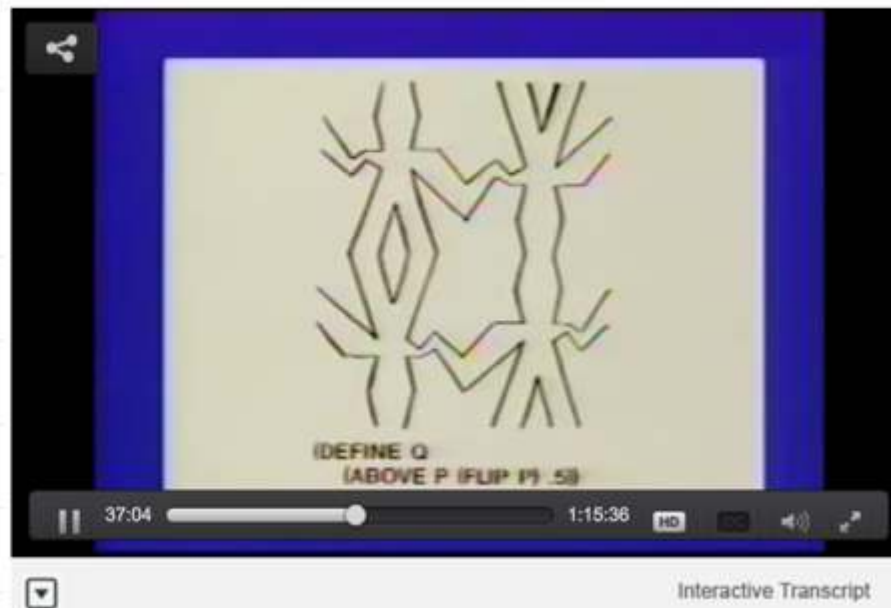
Peter Henderson
Functional geometry (1982, 2002)



Inspiration for this talk



SICP videos

3A: Henderson Escher Example[COURSE HOME](#)[SYLLABUS](#)[CALENDAR](#)[READINGS](#)[LECTURE NOTES](#)[VIDEO LECTURES](#) <[RECITATIONS](#)

F# inside

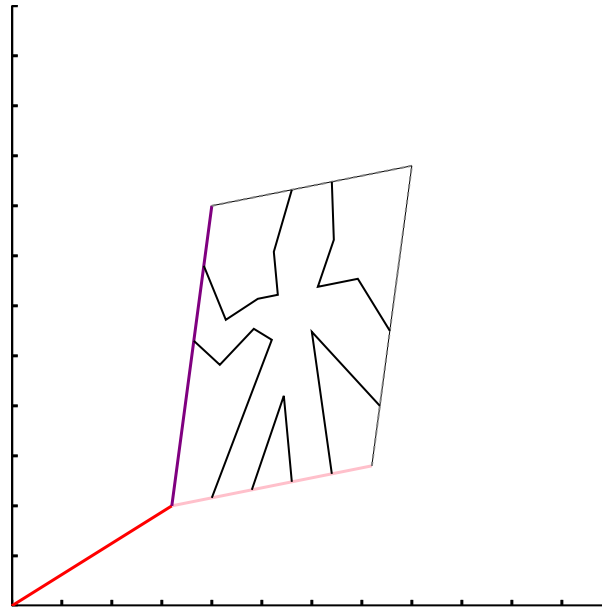
sfe

abstraction



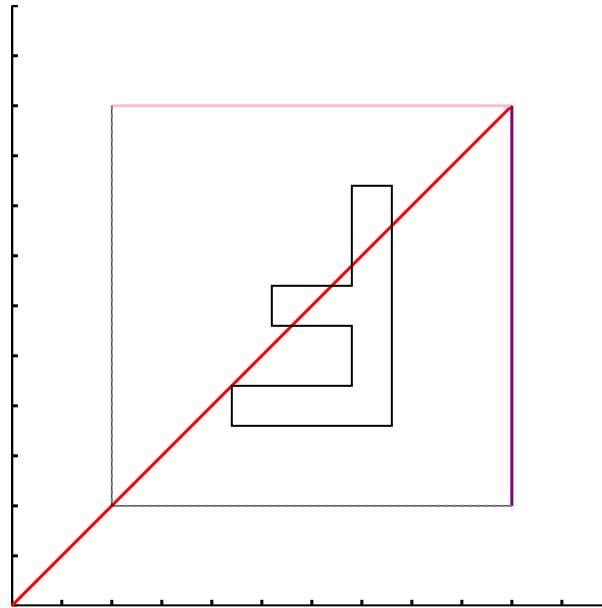
picture

type Picture = Box -> Rendering



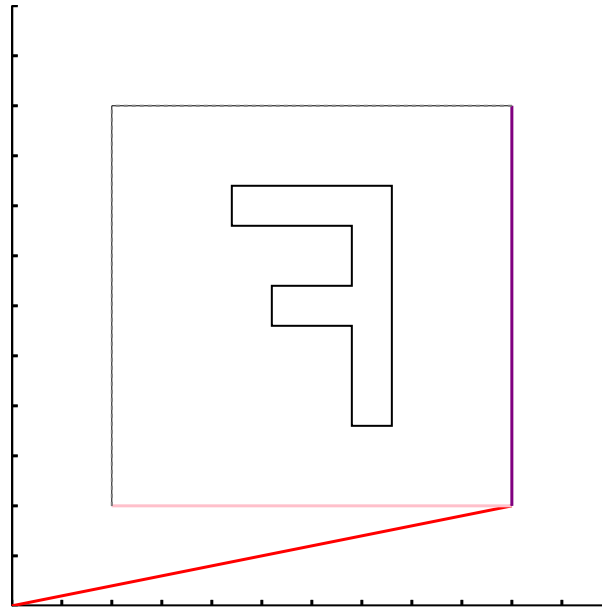
turn

$$(a', b', c') = (a + b, c, -b)$$



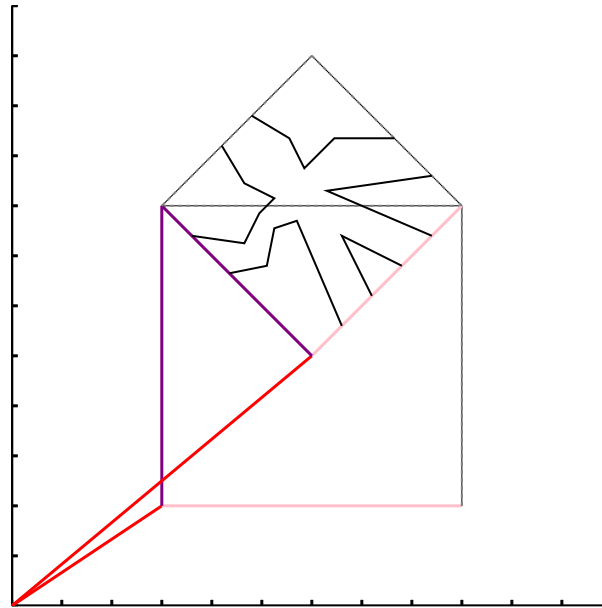
flip

$$(a', b', c') = (a + b, -b, c)$$



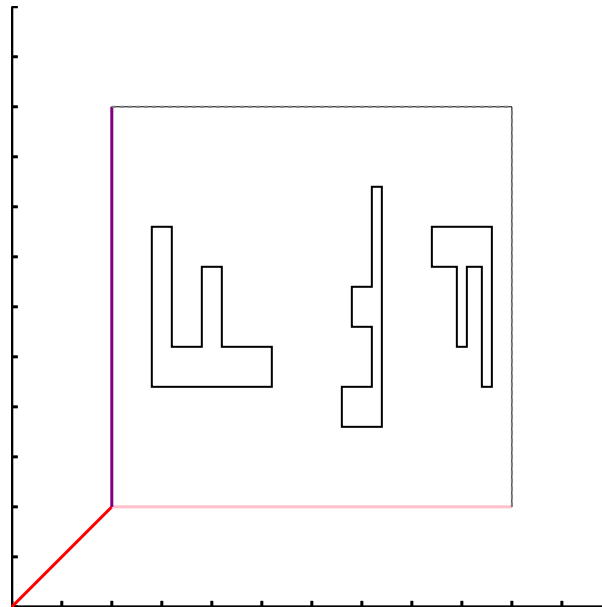
toss

$$(a', b', c') = (a + (b + c) / 2, (b + c) / 2, (c - b) / 2)$$



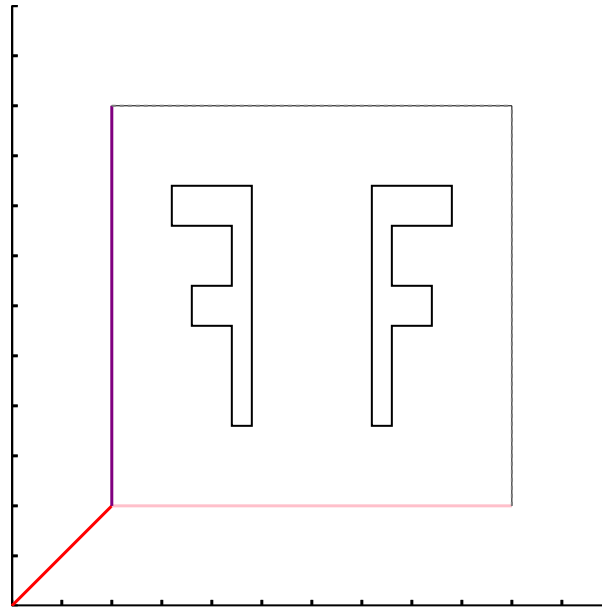
above

put first picture above second picture



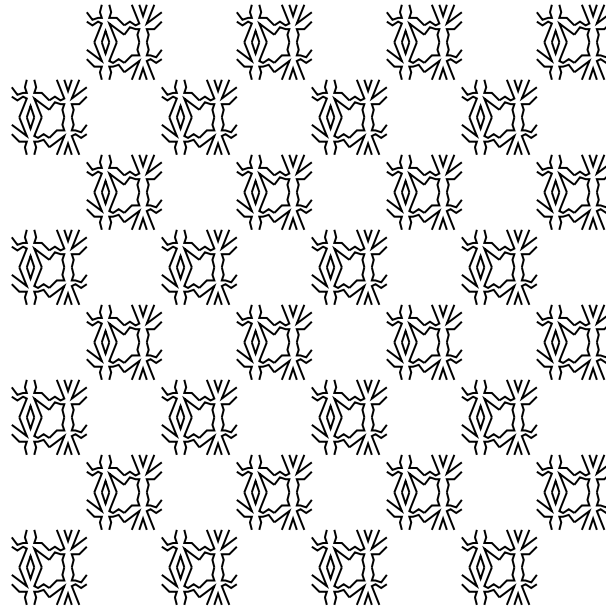
beside

put first picture to the left of second picture



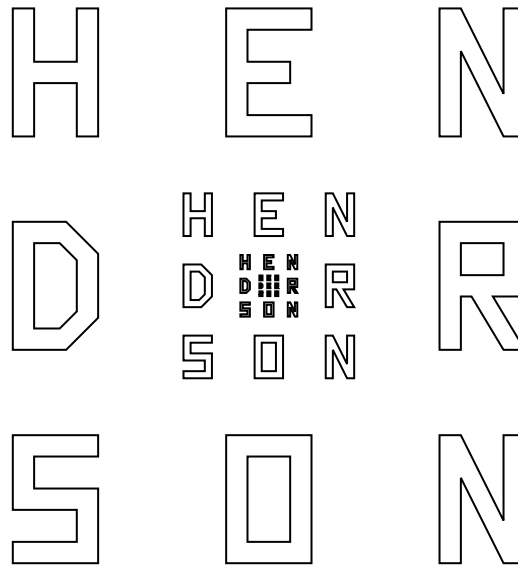
quartet

create a quartet of four pictures



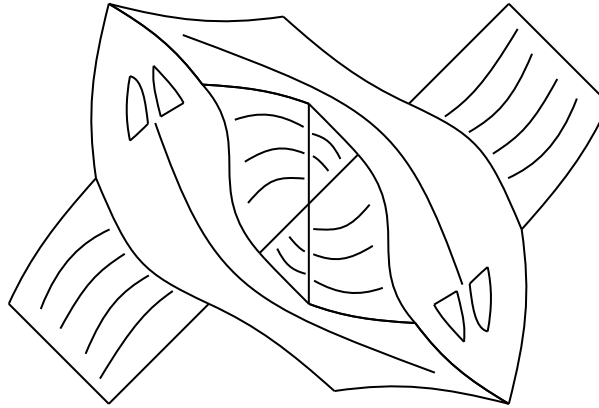
nonet

create a nonet of nine pictures



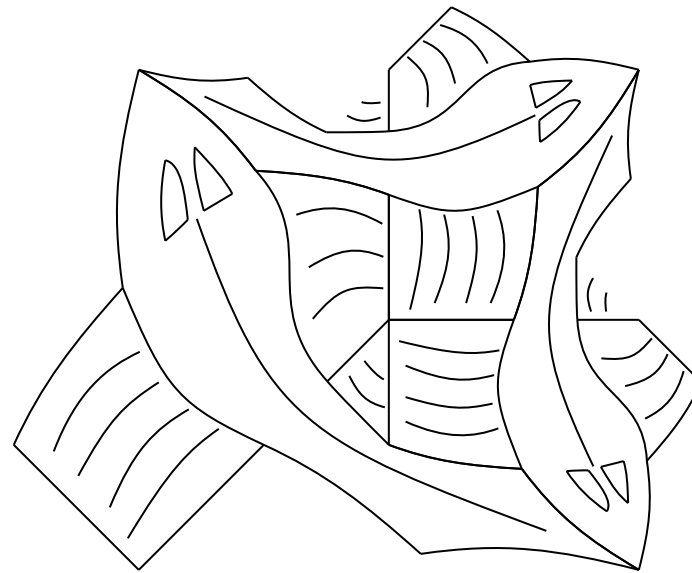
over

overlay two pictures inside the same box



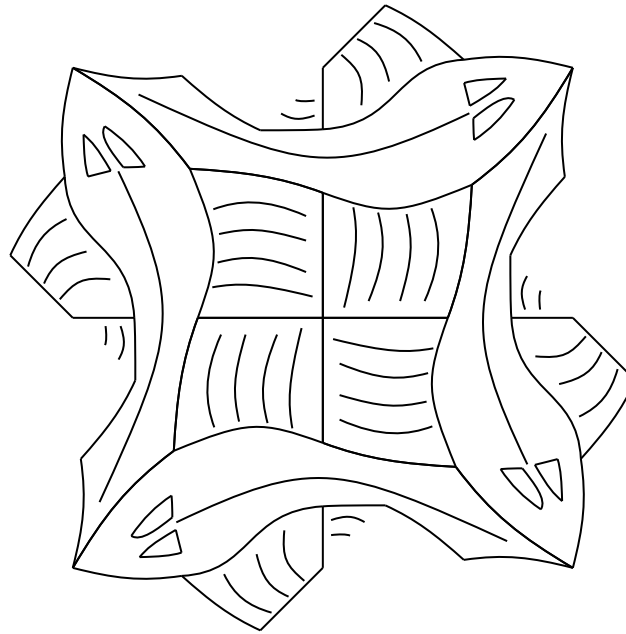
ttille

create the t-tile in square limit

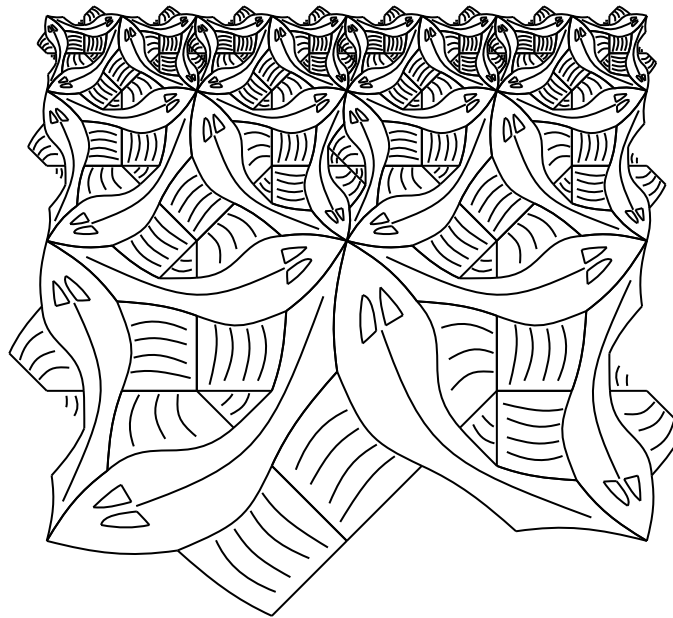


utile

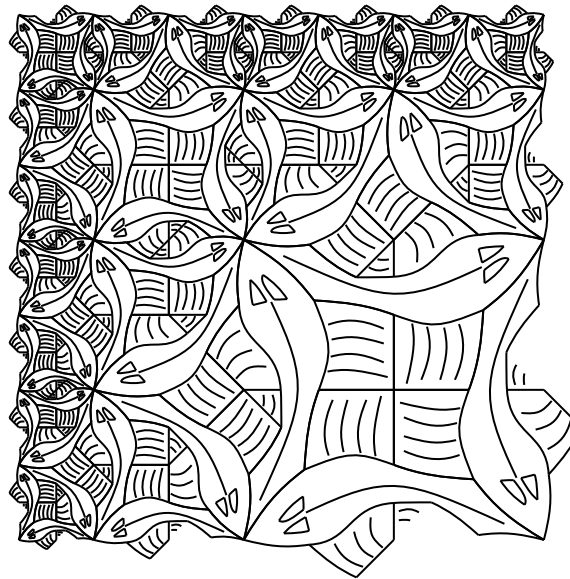
create the u-tile in square limit



side

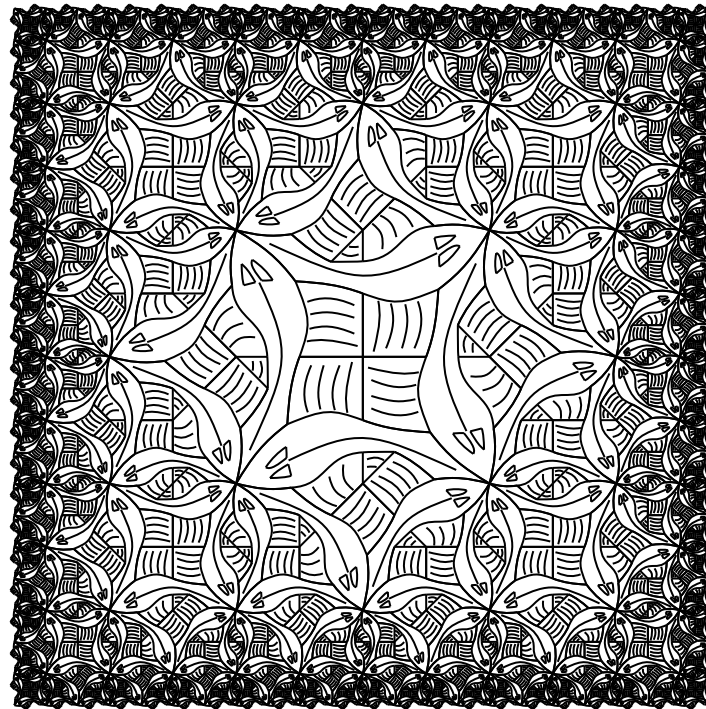


corner



square limit

Henderson's replica of square limit



hue



square limit

